

Math Workout

Variables and calc

A rectangle is 12 cm wide and 5 cm tall. Area: $12 \cdot 5 = 60 \text{ cm}^2$.

Double of 3,5 is 7.

Thousands still work: 100001.

Fractions, roots, powers

$$\frac{2}{3}$$

$$\frac{x+1}{x-1}$$

$$x^2$$

$$(x+1)^n$$

$$\sqrt[2]{\{x\}}$$

$$\sqrt[3]{\{x+1\}}$$

$$\sqrt[3]{\{27\}}$$

$$\sqrt{16}$$

$$|x-5|$$

$$\text{Nested: } \frac{\sqrt{4}}{2^3}$$

Trigonometry and logarithms

$$\sin(x)$$

$$\cos(x)$$

$$\tan(x)$$

$$\cot(x)$$

$$\sec(x)$$

$$\csc(x)$$

$$\arcsin(x)$$

$$\arccos(x)$$

$$\arctan(x)$$

$$\sinh(x)$$

$$\cosh(x)$$

$$\tanh(x)$$

$$\coth(x)$$

$$\log(10)$$

$$\ln(2.718)$$

Sums, products, limits

$$\sum_{i=1}^n(i)$$

$$\prod_{i=1}^n(i)$$

$$\lim_{x\rightarrow 0}\Big(\frac{\sin(x)}{x}\Big)$$

Integrals

$$\int x^2$$

$$\int x^2\,\mathrm{d}x$$

$$\int_0^1 x^2$$

$$\int_0^1 x^2\,\mathrm{d}x$$

$$\phi_C^F\,l$$

$$\int_0^1 x^2$$

Piecewise functions and equation systems

$$\begin{cases} x+y=3 \\ x-y=1 \end{cases}$$

$$\begin{cases} x x>0 \\ 0 x\leq 0 \end{cases}$$

$$\begin{cases} x x>0 \\ 0 x\leq 0 \end{cases}$$

Vectors and binomials

$\overrightarrow{AB}$

$\binom{n}{k}$

$\binom{5}{2}$

Matrices and tables

$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$

$\begin{pmatrix} a & b \\ c & d \end{pmatrix}$

Name	Age
Alice	20
Bob	22

Inline math and subscripts

The solution is $x = 2$.

Discriminant: $b^2 - 4ac$

Corners A_1 and x_n , vector $\overrightarrow{AB}$.

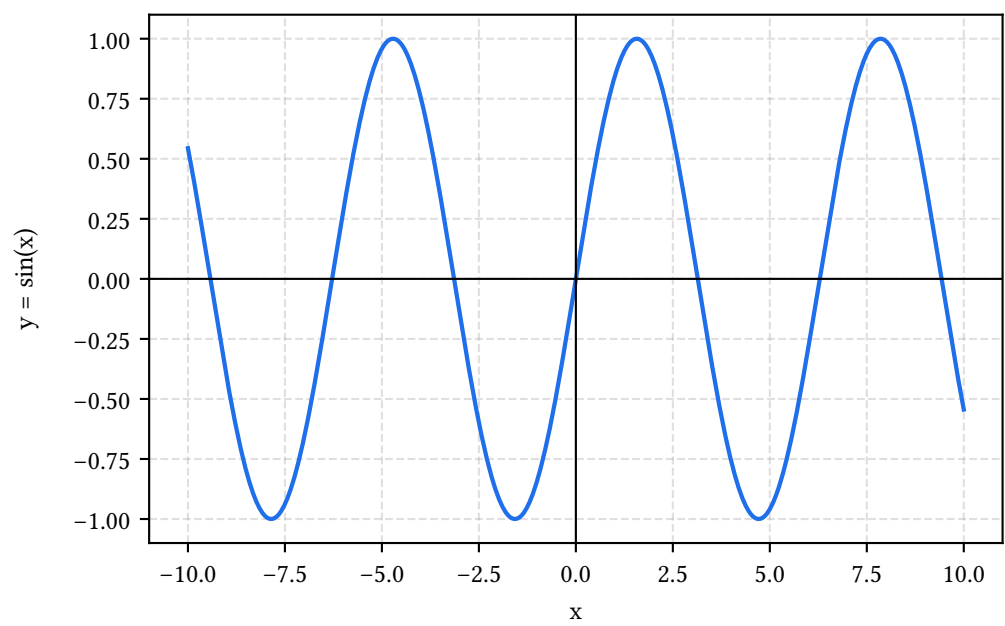
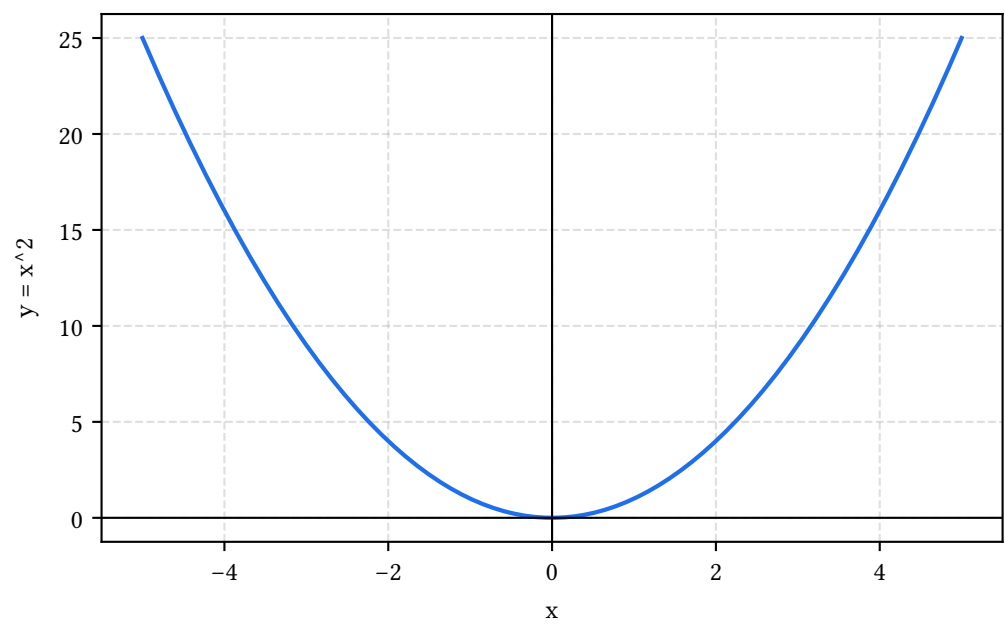
Spacing and symbols

a quad b quad quad c thin d med e

$\alpha \beta \gamma \pi \infty$

If $x \geq 0$ and $x \leq 10$ then $x \neq 5 \Rightarrow$ done.

Plots



Everything combined

$$\frac{\int_0^3 x^2}{\binom{4}{2}}$$